



Determining the Feasibility of Using Computer Vision for Visitor Use Management Data Collection

An Interactive Qualifying Project Report submitted to the Faculty of
WORCESTER POLYTECHNIC INSTITUTE
in partial fulfilment of the requirements for the
degree of Bachelor of Science

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Date:

29 July 2026

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This proposal represents the work of WPI undergraduate students submitted to the faculty as evidence of a degree requirement. WPI routinely publishes these reports on its website without editorial or peer review. For more information about the projects program at WPI, see <http://www.wpi.edu/Academics/Project>

Abstract

Acadia National Park wants to automate data collection and analysis via cameras to help make changes that improve the visitor experience. Cameras provide the data the park needs, but analyzing the images they produce by hand takes hours. Our goal was to determine if computer vision can process the image data that the park needs to reduce the time it takes to gather insights. We worked with staff and research teams to design our system and test it against field data. We were able to show that computer vision can be used to accurately calculate direction, speed, dwell time, and population of people, cars, and bikes. We recommend that our system be used to inform future development of computer vision analysis tools and that a customized camera system is built to support those tools.

Executive Summary

Determining the Feasibility of Computer Vision for Visitor Use Management Data Collection

Aditya Shah, David Bradbury-Roy, Elliot Scher, Kyle Plosky

Introduction & Background

Acadia National Park follows a dual mandate: it both preserves the natural beauty of the world and allows visitors to enter natural spaces for recreational purposes. To manage the park effectively and support the dual mandate, the National Park Service relies on visitation data to justify infrastructure investments, yet Acadia National Park's current monitoring tools are costly and insufficient to address the issues facing the park. Traditionally, this data would be collected by hand, which consumes staff time that could be spent on more worthwhile activities. Computer vision allows computers to analyze images like humans, transforming visual data into quantitative statistics without needing human involvement. Our goal was to determine the feasibility of using computer vision to process data and ease staff workloads.

Methods

We had three major tasks to support the park. First, we interviewed the park staff and performed participant observation with the other research groups at the national park to identify the requirements of the system. Following that, we placed cameras on-site at the Jordan Pond North parking lot, Bass Harbor lighthouse, Schoodic Peninsula, and along the carriage roads. We developed algorithms to enable the park to automatically count entities, determine occupancy in areas such as parking lots, compute dwell time of vehicles, determine the direction of travel of entities, and estimate the speed of entities. Using the data from the cameras we placed, we benchmarked these algorithms to see how accurate they were.

Findings

A Versatile System is Required to Meet the Park's Varied Needs

During our interview with the park staff, we discovered that the park avoids using cameras due to the extensive time required for staff to manually review footage. Most of their data is gathered using automated tools like car counters and Eco-Counters or through on-site manual procedures, with cameras used only when necessary. We also wanted to achieve feature-parity with their existing systems, which can determine counts, direction, and speed reliably. We consulted with the other research groups to identify the specific information they need and prioritized the most important aspects for our algorithm development, outlined in Table A.

Table A:

Table of features each group was interested in

Statistic	Park Staff	Jordan Pond	Schoodic Peninsula	Bass Harbor	E-bikes
Entity types	✓	required	required	required	✓
Counting	✓	required	✓	✓	✓
Direction	required	required	required	required	✓
Occupancy	✓	✓	✓	✓	
Dwell times	✓	✓	✓	✓	
Speed	✓		✓		✓
Clustering	✓			✓	
E-bike identification	✓				✓

Note. Check marks indicate that the group was explicitly interested in the statistic;

"required" indicates statistics that the group did not have a specific interest in but must be gathered to create the statistics they are interested in.

Specialized Camera Features and Placements are Required for Good Data

In our initial tests, we configured the cameras to take pictures only when detecting motion to avoid empty shots, but built-in and unpredictable delays made this method infeasible. When that did not produce the results we wanted, we started testing with video cameras; these had better results since continuous videos captured almost everything that passed by the camera.

We found that placing cameras perpendicular to the road was ideal for measuring speed and direction, while a 45° angle to the road was necessary to allow cameras to see vehicles' license plates for calculating dwell times. We decided to use 3-megapixel resolution as a balance between image quality, power, storage efficiency, and consistency. Lower resolutions resulted in less accurate detections. We also noted poor battery performance; cameras could only last about five hours with continuous video recording, and cellular cameras struggled for power even when solar-powered and set to take intermittent photos.

Computer Vision Meets the Needs of Acadia National Park

The results of testing our algorithms are summarized in the following table. We were able to identify at least one algorithm that showed promise in being able to provide usable data for each our goals. Table B shows a list of every algorithm that we created, the goal it addresses, and a simplified efficacy metric for that algorithm.

Table B:

Overview of algorithms, their purpose, and their efficacy

Goal	Algorithm	Media Type	Efficacy*
Entity Identification	Entity Classification	Photo	89%
		Video	
Direction	Pose Estimation	Photo	94%
		Video	94%
	Frame Track Analysis	Video	>99%
Occupancy	Overview Detection	Photo	87%
	Exit/Entry Detection	Video	98%
Dwell Time	Vehicle Reidentification	Photo	<25%
		Video	
	License Plate OCR	Photo	93%
		Video	
Speed	Frame Track Analysis	Video	93%

Note. Each algorithm was preprocessed with a YOLOv26 model to isolate entities of interest for further processing.

* The exact metric of efficacy varies between each algorithm and are not directly comparable.

Recommendation & Conclusion

We recommend that the park invest in specialized camera hardware that is reliable for video capture, real-time connectivity, and onboarding processing. Remote real-time data collection should be the next step, and proper hardware is important to have, since the hardware we used had several issues that need to be resolved for reliability. Ideally, the cameras should be able to upload high-resolution images or videos for real-time analysis and run our algorithms so they can upload a much smaller statistical report.

We recommend that the park create more algorithms for specific tasks like differentiating between e-bikes and traditional bikes. E-bike detection is challenging due to the lack of existing computer vision algorithms but is a worthwhile step if e-bikes gain more popularity and continue to be a topic of concern for the park. However, there are existing solutions that can be drawn from solving this problem and creating a versatile data collection system to help the park for years to come.

We recommend that the park use the tool we made and continue its development so it can be improved according to the park's needs. Using the software will give future developers and the park a better understanding of both groups' needs and allow for more effective development of computer vision tools in the future.

Our open-source system should be exhaustively tested by future groups. We did not gather information during inclement weather, which affects the credibility of our efficacy metrics. For future work, we should test algorithms with data from various weather conditions and explore more park areas to improve dwell time and clustering algorithms, making the system more representative.

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Appendix C: Observation Protocol	Elliot Scher	All
Appendix D: Speed Estimation Camera Orientation Comparison	Elliot Scher	All
Appendix E: User Manual	Kyle Plosky	All
Appendix F: Benchmarking Procedures	Elliot Scher	All

Meet the Team

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From left: Elliot Scher, Kyle Plosky, Aditya Shah, David Bradbury Roy

1. Introduction

Twenty years ago, national park staff would go out into the wilderness, armed with nothing but patience and a tally counter, and spend all day counting the people enjoying the beautiful landscape to get information the park needed to improve. These methods were time-consuming and inefficient, yet essential for efficiently managing the park's limited resources. Today, sensors collect visitor counts, and staff spend less time in the field counting and more time analyzing and acting on the data they collect. Technological solutions have become increasingly important as the number of visitors to the park has grown, and advanced computer vision technology has the potential to reduce the workload even further and keep up with the increase in visitation the park has seen in recent years.

National Park Service planning policy recommends using visitation statistics to justify infrastructure changes and management decisions needed to accommodate increased visitor numbers (National Park Service, 2019). Acadia National Park's current monitoring techniques are expensive and provide insufficient information to address challenges promptly. Computer vision is a promising approach for improving camera-based visitor use management (VUM) systems at Acadia National Park.

Our project assesses the feasibility of using computer vision to ease analysis of photographic data in Acadia National Park. Computer vision uses complex algorithms to help computers interpret images and videos by recognizing patterns in objects, people, and scenes (Waelen, 2023). We start this report by covering the importance of VUM and how computer vision can be applied to VUM data collection tasks to help with the overcrowding situation at Acadia National Park. Next, we describe the implementation of a proof-of-concept management system and the factors that went into designing and testing it. Finally, we present our findings and provide recommendations for continued development.

2. Background

This chapter discusses how computer vision can be utilized to aid in VUM data collection tasks. We consider the importance of VUM techniques and how they are used by national parks, and how computer vision might fit into VUM systems. We also discuss the importance of training data and how to create a system that considers privacy, ethics, and sustainability in Acadia National Park seriously, then explore the specific considerations that need to be made for Acadia.

2.1 Visitor Use Management

Visitor use management, or VUM, is a framework that guides national parks in supporting their dual mandate of preserving natural spaces while providing recreational access to them. National parks employ VUM techniques to understand how visitors use the space and improve it accordingly (Interagency Visitor Use Management Council, 2019). VUM allows parks to manage crowds and environmental issues alike by informing staff dispatch, signage placement, and the determination of areas to renovate or reintegrate into nature.

An important part of the framework is data collection, which serves as evidence to support budgeting and inform decision-making leading toward VUM's goals. This data often consists of occupancy trends of how many people go to certain locations, dwell time data of how long they stay there, and the direction that they travel through the area (Interagency Visitor Use Management Council, 2019). This helps national parks protect the environment while also accommodating visitors' needs. For example, the data may reveal locations where visitors are creating their own trails through a park and causing erosion or disturbing wildlife (Manning et al., 2006). It can also help determine the maximum number of people that can safely and comfortably occupy an area (Viñals et al., 2024). Without these metrics, it may be difficult to justify funding or policy changes necessary to protect natural resources.

The most basic form of data collection consists of manual observations from park staff (Watson et al., 2000). While extremely simple to manage, these techniques require

human park staff to be present wherever monitoring needs to occur, making this method costly and time-consuming. Human observations also leave some room for bias and variance if a detailed observation protocol is not defined (Jones et al., 2018).

Technological improvements in the past few decades have enabled the development of automated counting devices like break-beam trail counters and weight-based car counters (Watson et al., 2000). These devices are effective and simple, but they can be easily triggered by wildlife and require intrusive construction work to be installed on trails. Additionally, these sensors only count the number of visitors and may be unable to provide more detailed insights, such as the specific activity that a visitor is engaging in (Jones et al., 2018).

2.2 Computer Vision

Cameras can be used to collect more detailed information than automated counting devices without requiring constant observation, but they traditionally require manual review to extract usable data (Lupp et al., 2021). Researchers have explored using computer vision to remove the step of manually reviewing images one by one (Albers et al., 2023; Miguel et al., 2024). Computer vision algorithms allow computers to “look at” images as humans do and convert qualitative information in the form of images into quantitative statistical data without requiring human input. Computer vision systems have been proven to accurately classify animals (Zuolkernan et al., 2022), vehicles (Tian et al., 2015), and even visitors to national parks and other outdoor spaces (Albers et al., 2023).

Modern computer vision systems use machine learning to accurately analyze images. Machine learning is a field of data science where computers learn to analyze data without the help of humans by using a large set of examples, called a training set. Once trained, the computer can perform the task it learned with information that was not in the training set (Louridas and Ebert, 2016).

Computer vision systems often consist of multiple layers that analyze the same data in a sequence, drawing from the results of the previous layers. For example, a model looking to track which way people go at an intersection could start with a quick and

efficient layer for locating people in the image, with further layers using that data to determine which direction the people are moving or match the appearance of a single person across multiple images (Liu et al., 2024). Layering processes like this example can lead to complex insights from simple models.

The first, foundational step in layered computer vision is entity detection. An entity is a specific object of interest; usually a person, car, or bike for this project. Performing entity detection on an image provides us with a list of tagged boxes, each of which is associated with a single entity. The bounding box of an entity is the tightest box around the entity possible, and the tag specifies which type of entity was found. There are many systems that already exist for detecting entities, including the YOLO family. YOLO models are accurate, relatively small, and can be run on ordinary computers and embedded systems, making them a good choice for this task (Jocher et al., 2026).

Determining visitor movement direction is an important metric that can be used to find routes visitors use to move through the park as well as assist in determining occupancy. Time-based direction works by finding an entity in two separate images at different times and figuring out how their position changed, then using that movement to find their travel direction (Monika et al., 2023). In high-framerate video, entities can usually be tracked based on how much their bounding boxes overlap, but in lower framerates and across images that have seconds between them, other complex reidentification techniques need to be used (Butoto et al., 2026). Instead of using time-based models, a specially trained machine learning model can also be used to identify the direction a person or vehicle is facing from a single image based on the pose of its parts relative to each other, which works better in situations where capturing multiple images in quick succession is impractical (Zhao et al., 2012).

Dwell time—the measure of how long an object stays in a certain area—is another common metric for understanding visitor behavior and the popularity of a space. Calculating dwell time relies on a computer vision technique called reidentification—recognizing entities that have been seen before. To do this, computer vision systems must create a unique identifier for entities that pass through a camera’s view. The system will

then try to match the identifier against entities that pass through the camera's view again. When the match is successful and the directions of travel were different, the system can use the two images as entrance and exit times to calculate dwell time. The simplest way to identify vehicles is to use license plate information as the identifier; license plate numbers can be extracted from high-quality images using optical character recognition (OCR) (Lu et al., 2025). OCR can occasionally misread characters, but this can be mitigated using a method called "fuzzy matching," which matches text even if a few characters are missing (Singh and Singh, 2025). Alternatively, a mathematical representation of the objects in an image can be created by analyzing the image in small sections. These pieces are a unique "fingerprint" of what a specific object looks like and can be used as an identifier instead of license plates (Butoto et al., 2026). Either of these methods can be used to reidentify objects that come back through a camera's field of view, and the difference between the timestamps from both images can provide dwell time information.

Occupancy data—the number of people in an area over time—can be used to understand the balance of visitors across the managed area and to find areas which are likely to need more careful attention. There are two main ways to measure occupancy: by monitoring an entire area with an overview camera, or by monitoring all the entrances and exits to an area. The occupancy of parking lots and other areas with specifically allocated "seats" is often measured by identifying entities in an overview image and figuring out which seat they are in (Ahad et al., 2026). Occupancy in other areas without specific slots can also be measured with this method simply by counting the number of entities in view of the camera. Entrance and exit monitoring is an alternative strategy which works best in situations where it is impractical to have a camera or set of cameras that can see an entire area. This technique requires cameras to be placed on the entrances and exits to an area and will update a counter whenever entities are detected entering or exiting the area, based on their direction of travel (Arshad et al., 2022).

Measuring the speed of objects is foundational to learning about how the park is used and how that use is affecting the park's natural resources. Speed information can be difficult to capture because two sequential frames with a known time difference are

required to calculate speed. This method is only useful for calculating relative speed, as there is no frame of reference for converting pixel lengths to real-world lengths. To find exact speed information, there must be a known conversion factor from pixels into units of length, accomplished using either an object of known shape and size in the background of the images or an object travelling at a known speed (Yohannes et al., 2023).

Visitors behave differently when they are alone and when they are in a group. Because of this, identifying groups of entities can be valuable when assessing visitor behavior. Some complex algorithms detect and count individual people within groups by systematically extracting each person and finding others who are most closely associated with them, using indicators like texture, edge density, or pixel ratio (Bai et al., 2021). A simpler method compares the distance between different entities and groups those that are near each other. Additional heuristics can be used to increase accuracy. Once individuals are put into cluster groups, they can be counted to determine group sizes.

2.3 Data Considerations

The accuracy of machine learning systems depends on the quality of the data with which they are trained and the quality of the data input. Using incomplete or incorrect information with machine learning models can result in unreliable predictions (Mohammed et al., 2025). Because of this, it is important to check how accurate the data is to make sure that any conclusions reached from the model are based on trustworthy information. With large quantities of data, a preprocessor can identify and remove outliers for better processing. A statistical model looking at patterns in images can help detect poor-quality data and remove outliers in a set of images, leading to more consistent results (Nadadur et al, 2005).

Privacy issues and ethical concerns need to be carefully considered when collecting data for and using machine learning and computer vision. Clarity about data use and user privacy is needed to build trust and responsibility in the use of surveillance technologies (Li, 2023). Machine learning systems can reinforce the social biases present in their training data, which can lead to unfair treatment in crucial areas of life. This can be

improved by using a wide variety of training data that represents diverse groups of people (Li, 2023). In addition to ethics and privacy, sustainability is an important factor to consider when developing a machine learning system. Training large-scale models emits greenhouse gases comparable to the emissions of five cars during their entire lifetime (Wang et al., 2025). Given these environmental concerns, it is crucial to prioritize sustainable principles in system design.

2.4 VUM in Acadia National Park

Acadia National Park sees around four million visitors annually, placing it among the top ten parks in the country in terms of yearly visitation (National Park Service, 2025). Despite its popularity, the park sits on just under 50,000 acres along the coast of Maine, significantly smaller than the sizes of similarly popular parks (National Park Service, 2026). This is further constrained by the region's short tourist season, putting most of the guests within the span of a few months per year (Fleming, 2016). With visitation projected to continue to increase, Acadia National Park is suffering from the effects of overcrowding and is looking to make changes to maintain the quality of visitor experiences and the natural environment.

In response to these concerns, Acadia National Park has developed a transportation plan intended to help manage congestion by modifying the park's transport infrastructure in line with the VUM framework (National Park Service, 2019). It outlines several metrics for determining if infrastructural changes to the park are serving their purpose. Park management strategy recommends using automated data collection as a means of gathering this information at a large scale but does not describe the specific systems needed to do so (National Park Service, 2020). Acadia National Park already has various automated data collection tools installed around the park. Car counters have been set up to track the congestion of cars on main routes in the park and can record the time, speed, and direction that vehicles pass by them. Eco-Counters have been installed on the trails and carriage roads to count passing pedestrians and bikers and provide the time and direction of each crossing. While they require less effort than manual counting, these

methods still require that a person visit each device to collect its data and provide significantly less detailed information.

Acadia National Park management also makes use of computer simulation and modelling to predict and approximate visitation over time (Manning et al., 2006). These models provide rough estimates of visitor counts throughout the park, but they still need to be backed by real data, particularly in times when visitation patterns are changing. The transportation plan suggests future work be done to improve the data collection systems that back up these models (National Park Service, 2019). With congestion on the rise, making these improvements is a necessary upgrade to the park's infrastructure.

The congestion problems in Acadia National Park demonstrate the need for an improved VUM data collection system. Recent advances in computer vision and machine learning provide the opportunity to collect visual data from cameras and turn it into numerical data suitable for analysis. This project intends to determine the feasibility of using computer vision to manage visitors in Acadia National Park by developing and evaluating a prototype data collection system. During the process of developing such a system it is important to assess how well it is fulfilling its purpose compared to other data collection techniques and care must also be taken to ensure that the system conforms to the privacy and ethical standards of the park.

3. Methodology

This project aimed to help Acadia National Park and Friends of Acadia process visitor use management data more effectively with the addition of computer vision. To achieve this goal, we identified the following objectives, which we discuss in-depth in this chapter:

1. Identify the data needed by the park and other research teams for effective visitor use management.
2. Determine optimal camera configurations for collecting accurate statistics.
3. Develop a computer vision pipeline and compare it to other data collection methods.

3.1 Objective 1: Visitor Use Management Analysis

Before developing our computer vision model, we needed to learn what data the park uses for VUM so we could design, train, and test an effective solution. We interviewed park staff and performed participant observations with other research groups to determine key features of VUM data collection, including the current methods and the purpose, capabilities, and limitations of each method. This helped to inform where to set up cameras and what kinds of data our system needed to process.

Before leaving for the project site, we conducted a semi-structured expert interview with the park staff to learn about their current VUM practices. We interviewed Becca Stanley, recreation technician manager at Friends of Acadia, and park rangers Adam Gibson and John Kelly to learn about how, where, and when they decide to use cameras in park-organized projects. The interviews followed the interview guide in Appendix A, while also using probing questions to follow up on relevant information and extend the conversation over topics we did not foresee discussing in our guide (Berg & Lune, 2017). The interviews were analyzed with grounded theory, where we used *a priori* ground truth to contextualize and categorize the information we gained from these interviews, with appropriate adjustments made as new themes emerged in the data (Berg & Lune, 2017).

We were unable to record the interview, but we took notes extensively and had a discussion of the topics covered immediately after the interview was completed.

Additionally, we performed participant observations to gather input from other research groups working at Jordan Pond, Schoodic Peninsula, the Bass Harbor Head Lighthouse, and on e-bike usage on the carriage roads, which allowed us to quickly and conveniently gather information about how computer vision can be used to meet the park's needs. These participant observations helped us understand the types of data they wanted to access and how they intended to collect that data from an insider's perspective. Our researchers took notes as they participated in the decision-making process of the other groups and provided their own insight.

In combination with the information from the park staff, we determined a set of metrics which would be useful for the park staff and future groups and seemed to be possible to analyze using computer vision. We used the information collected during this stage of research to inform our further work on our other objectives.

3.2 Objective 2: Camera Configuration and Data Collection

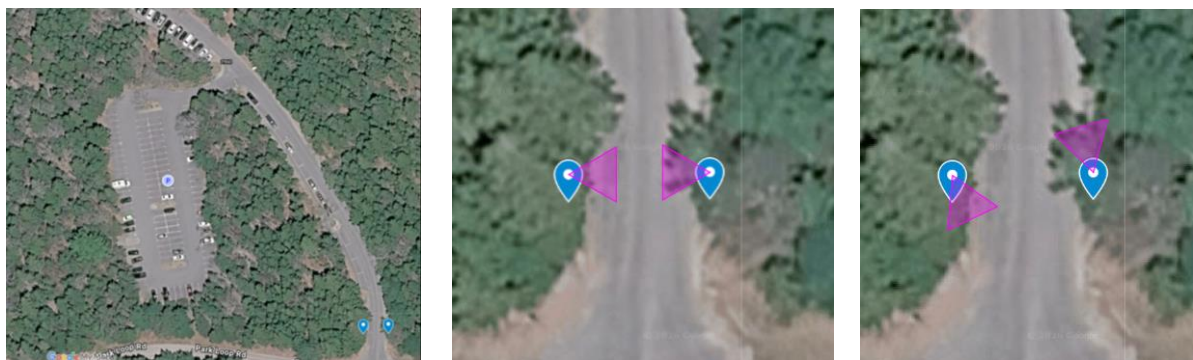
To evaluate computer vision's applicability to VUM data collection, we needed to collect images from various areas of interest around Acadia National Park. The other research group's requirements gave us a set of convenient locations where we already understood the types of data being collected. As such, we set up cameras at the entrance to the Jordan Pond North Lot, Bass Harbor Head Lighthouse, various locations along Schoodic Peninsula, and on the Eagle Lake carriage roads. Each location tested a unique part of our computer vision system.

The Jordan Pond research group was focused on dwell time and occupancy in the Jordan Pond parking lot to inform the park staff about a potential reservation system, so we used the photos and videos collected at this location to develop our models for occupancy and dwell time. To collect this data, two VOOPEAK TC02 trail cameras were set up oriented to face the single entry/exit point to the Jordan Pond parking lot, as shown in the middle photo of Figure 1. Occupancy was calculated from this position using the entrance and exit

monitoring technique. To calculate dwell time, the cameras had to be reoriented to capture the license plates of the cars entering and exiting the lot, as shown in the right photo of Figure 1. Dwell times were calculated by tracking the timestamp when a particular license plate entered and exited the lot. The cameras were tested with motion detection, time-lapse, and continuous video recording.

Figure 1:

Jordan Pond North Lot camera placements, overview, initial, and revised.



The research team at Bass Harbor Head Lighthouse aimed to collect occupancy and clustering data from various locations around the lighthouse at different times of day. We set up four cameras, each with an overview of key areas of interest around the lighthouse (Figure 2). All the cameras were Spypoint Force 20 cameras except for the one on the rocks, which was an AKASO TC04. We experimented with several camera positions and angles within these areas to optimize the field of view, as well as triggering the cameras using both motion sensors and timed intervals. We calculated occupancy at these locations by counting the number of people captured in overview images at regular intervals.

Figure 2:

Final Bass Harbor Head Lighthouse camera placements

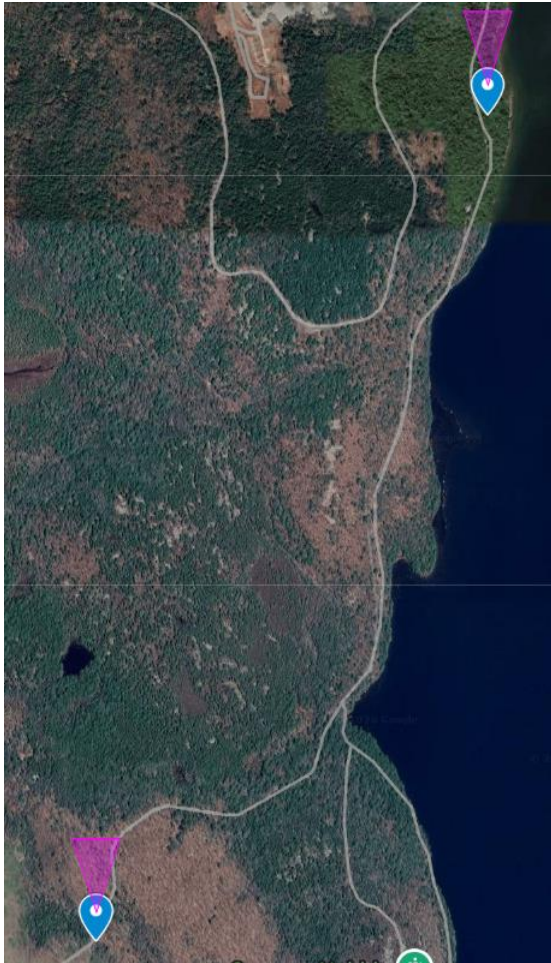


Note. The lighthouse overlook had not been built yet when this satellite image was taken; the camera on the overlook is marked at the location where the overlook has been built.

The research team focused on e-bikes was interested in collecting data about the different types of visitors using the carriage roads including their speed and direction of travel. We set up two trail cameras: an AKASO TC04 near an Eco-Counter for comparison to the Eco-Counter data, and a VOOPEAK TC02 on a hill to look for differences between the speed of bikes and e-bikes (Figure 3). We eventually realized that detecting e-bikes was out of scope, so the data from the hill counter was unused. We initially set the cameras to run motion detection to avoid empty images, but we switched to continuous video when the cameras failed to capture the images we needed.

Figure 3:

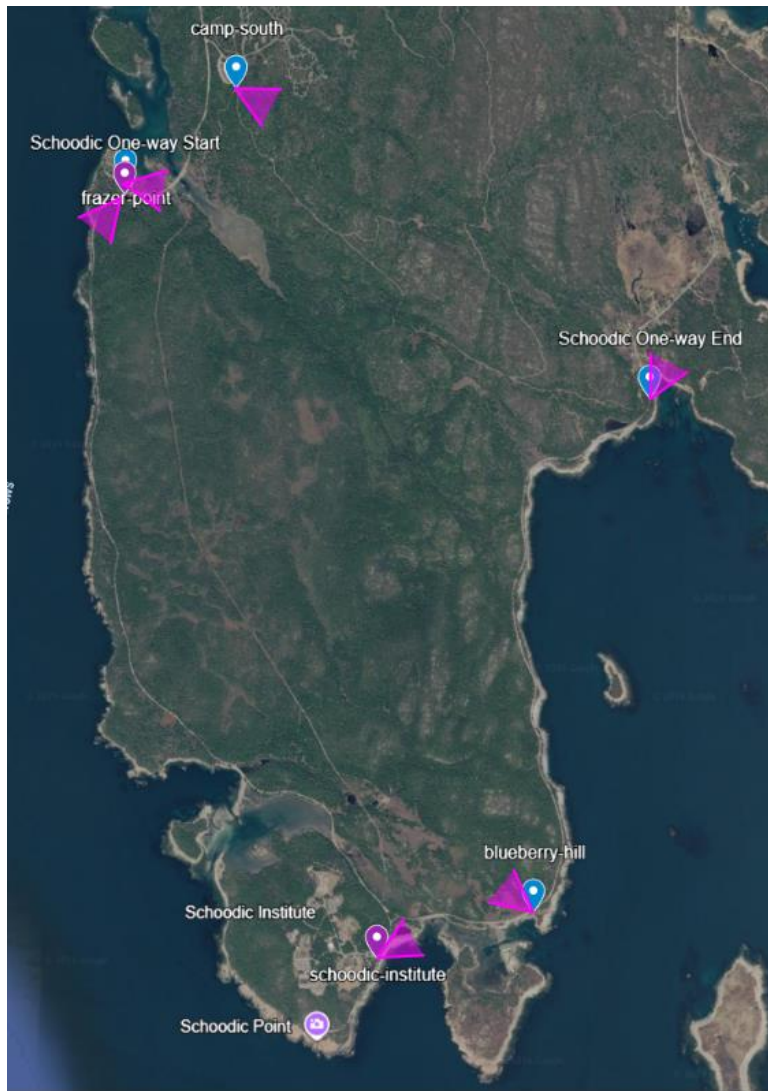
Carriage road camera placements on the west side of Eagle Lake



The Schoodic Peninsula research group wanted to understand the visitor experience on Schoodic Peninsula. Cameras would help the team calculate occupancy and dwell time in several parking lots and across the main one-way road. We placed a combination of cellular (Spypoint Flex-S Dark) and non-cellular (Spypoint Force 20 and AKASO TC04) cameras at strategic positions along the road and at parking lot entrances, as seen in Figure 4. Cellular cameras were placed at locations that were most impractical to access frequently, though ideally every camera on the peninsula would be cellular. The cameras at the start and end of the one-way were meant to determine overall dwell time and occupancy on the peninsula, while those at the parking lots were intended to calculate both the occupancy and dwell time of the people in those parking lots.

Figure 4:

Schoodic Peninsula camera placements



Note. The blue markers are non-cellular cameras, and the purple markers are cellular cameras. The Schoodic Institute camera is off the main loop, and the one at the start of the one-way section is at the roadside where there is no convenient place to stop.

Once all the cameras were placed, we checked them regularly to ensure that their batteries stayed charged and their memory cards did not fill up. We also made iterative adjustments to the camera angles and settings based upon the quality of the images they gathered.

3.3 Objective 3: Algorithm Evaluation

Once suitable training data was collected, we developed and evaluated computer vision algorithms designed to automate extraction of VUM-ready data from images. We collected images from cameras and transferred them to a database for further processing and analysis. All images were deleted at the conclusion of the study to preserve visitor privacy.

The first step of our evaluation process was to determine if the photos that our cameras captured were accurate representations of reality. To do this, we performed manual counts on the Witch Hole Pond Loop and on the east and west sides of Eagle Lake. All our counting was done next to Eco-Counters and with a camera set up next to us so we could compare our counts and camera data to reality and the park's existing counting methods. All of the images and videos collected from the cameras were manually processed to ensure that the images the cameras captured were accurate, and the counts from the cameras, the reality that we counted by hand, the data from the Eco-Counters, and the output of our algorithm were all compared side-by-side on histograms.

For most of our testing, we compared our algorithms to data collected by manually assessing photos from the cameras. This allowed us to assess our algorithm with data we collected while not present on site. To test algorithms which required precise timing, we used long-duration videos from phones and the trail cameras, which were nearly guaranteed to visually capture the data that we needed. We took videos next to the western Eagle Lake Eco-Counter and compared the data from it with the base model. We also gathered video data from Jordan Pond, which we manually labelled and used to test our occupancy algorithm and license plate detection for dwell times.

We created two different direction algorithms which had to be tested—one for video and one for images. The image algorithm was tested by manually labelling the direction of each entity in a set of images and comparing our algorithm's guesses to the labelled data. The video algorithm was tested with a similar method, only using video data as the source. To test our speed algorithm, we gathered videos of a street next to one of the park's car counters. The data on the car counter was not collected while we were on site, so we were

unable to use it as an exact ground truth. We were able to compare the car counter’s data to the historical data using the Jensen–Shannon divergence test, which provides a metric of how closely data distributions align with each other and is appropriate for comparing trends (Xue et al., 2025). More detailed benchmarking procedures are documented in Appendix F.

4. Findings

As we performed our methods, we recognized several key findings pertaining to the design of a computer vision data collection system in Acadia National Park. We organized these findings around three main themes: the specific needs of a data collection system, important camera selection and placement considerations, and the outcomes of our algorithms.

4.1 A Versatile System is Required to Meet the Park's Varied Needs

Our first objective was to learn about the data the park and other research groups use to inform visitor use management decisions. We gathered this data by interviewing park staff and performing participant observations with other research groups in the field.

In our interview with park management, we learned that the park avoids using cameras if possible because it takes many hours to manually review the images they produce. Most of the park's current VUM data collection is conducted using automated counting tools such as car counters or Eco-Counters; cameras are only used to collect data that cannot be collected by other automated methods. The park staff mentioned a study that looked at whether dogs in the park were on leashes or not as an example; the preexisting monitoring technology was unable to automatically calculate this statistic, so they used cameras and manually processed the images.

The park currently collects counting data from numerous locations across the carriage roads and hiking trails, as well as a few of the main roads through the park. They mentioned how their systems are unreliable when it comes to differentiating different entities that pass their counters. They also mentioned that they collect direction from those counters, and they usually have speed data available as well.

This helped us understand that the park's data collection needs—especially those involving cameras—are constantly changing. Since cameras are assigned to unusual data collection tasks that cannot be done using simpler tools, we determined that a computer vision data collection system would need to be flexible to meet the park's needs. We turned to our participant observations with each of the other research groups to

understand more specific types of information they want to gather and discover the most important ones to focus on through the development of our algorithm. These needs are documented in Table 1.

Table 1:

Feature matrix of groups and data needs

Statistic	Park Staff	Jordan Pond	Schoodic Peninsula	Bass Harbor	E-bikes
Entity types	✓	required	required	required	✓
Counting	✓	required	✓	✓	✓
Direction	required	required	required	required	✓
Occupancy	✓	✓	✓	✓	
Dwell times	✓	✓	✓	✓	
Speed	✓		✓		✓
Clustering	✓			✓	
E-bike identification	✓				✓

Note. Check marks indicate that the group was explicitly interested in the statistic; "required" indicates statistics that the group did not have a specific interest in but must be gathered to create the statistics they are interested in.

4.2 Computer Vision Requires Special Camera Placements and Features

During our initial tests, we set up the cameras to take pictures only when they detected motion to avoid capturing empty photos. This would provide us with sufficient information for analysis if the cameras consistently captured images when they detected motion. We soon found out that the photo frequency was too low to record large groups of visitors accurately. While manually reviewing the images before running the algorithm, it became clear that the issue lay with the cameras themselves, which were incapable of capturing data that represented reality. Our testing on the carriage roads revealed that when a large group of bikes passed, the cameras consistently failed to capture every bike in the group, resulting in severe undercounting compared to our manual counts. Additionally, the cameras had extremely inconsistent delays between photos and between detecting motion and taking the first photo. As shown in Figure 5, the camera captured 23% of the

349 bikes and 65% of the 23 pedestrians that were manually counted, indicating that capturing still images with motion detection would not yield usable data.

Testing using video cameras yielded much better results, as a continuous video stream catches almost everything that passes by and collects the data necessary to calculate speed and direction on top of standard detection and counting. While several cameras only provided unreliable short bursts or did not have this capability at all, we still managed to obtain some video data from the VOOPEAK and AKASO models. A comparison of several relevant camera features is presented in Table 2. Our testing with a video camera on the Eagle Lake West carriage road, shown in Figure 6, captured 97% of the 197 bikes and 127% of the 30 pedestrians that passed by. The camera captured more people than were counted in person because the camera's clock was behind by a few minutes, resulting in additional recordings after the manual counting had wrapped up.

Figure 5:

Comparison between counting methods at Duck Brook Bridge using photos

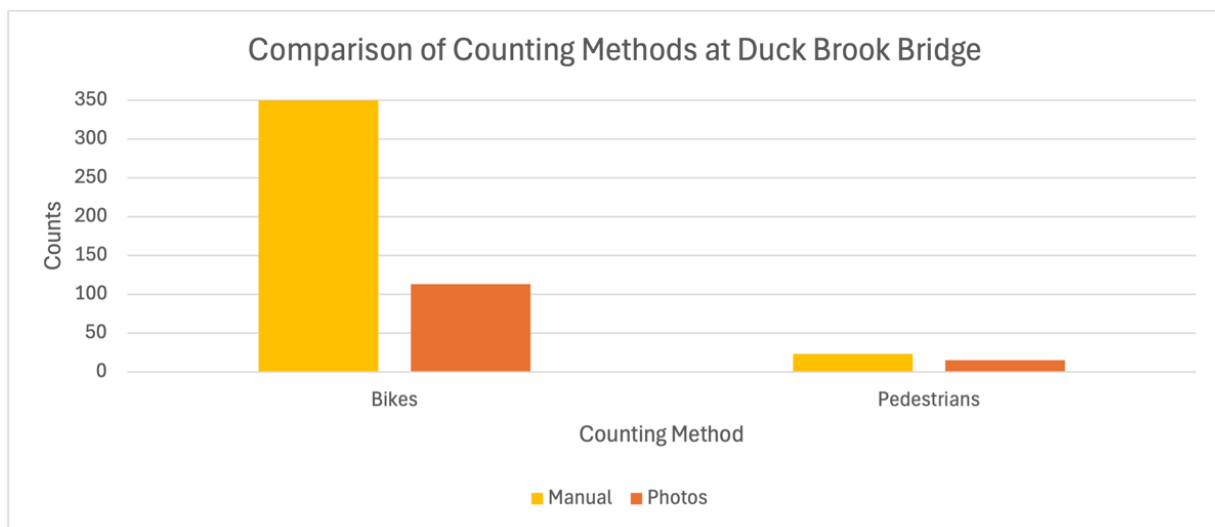
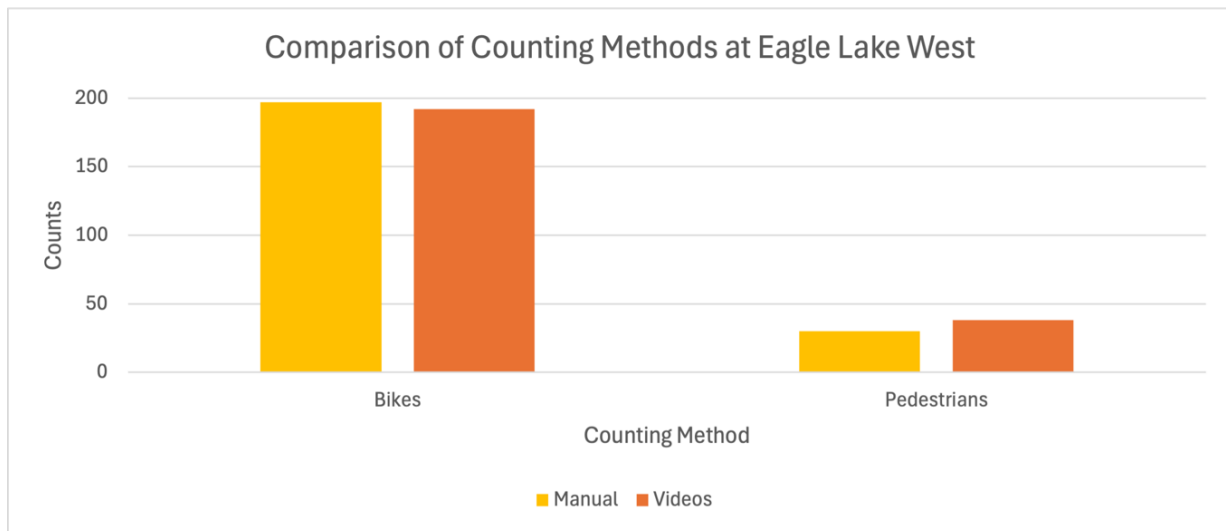


Table 2:*Comparing continuous videos and power of different cameras*

Camera	Continuous Videos*	Solar	Cellular
VOOPEAK TC02	✓	✓	
Spypoint Link-S		✓	✓
Spypoint Force-20			
Spypoint Flex-S Dark		✓	✓
AKASO TC04	✓		

*Continuous video is not a supported feature of any of these cameras; to enable it we had to enter setup mode and take a test video, then never stop recording. This mode is officially intended for very short uses to test the camera, not for recording hours of footage.

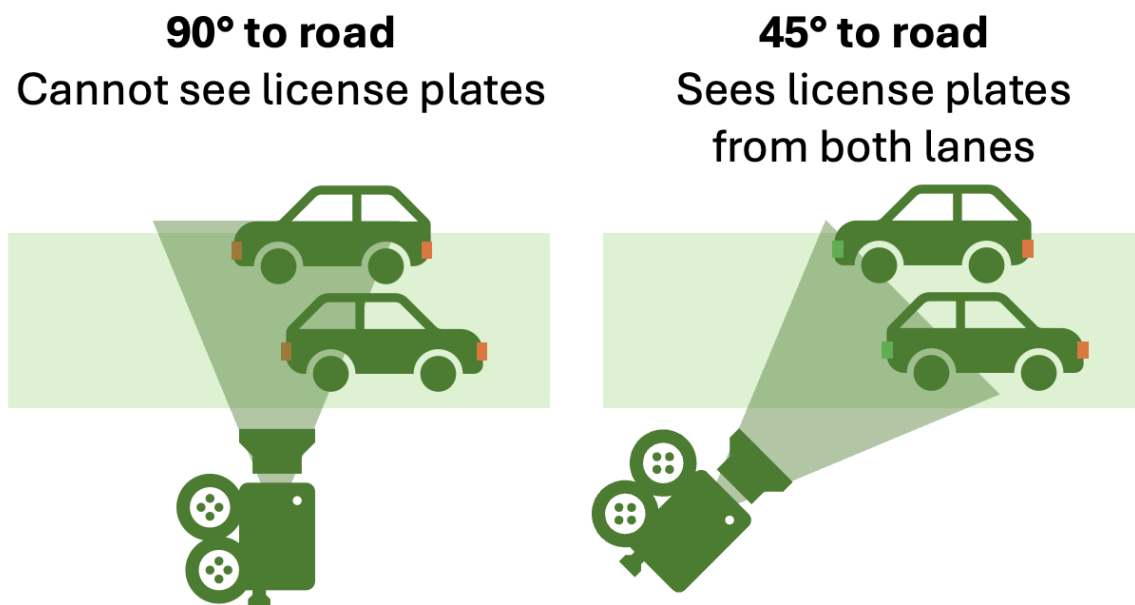
Figure 6:*Comparison between counting methods at Eagle Lake West using videos*

When placing cameras, it is important to consider the angle of the camera. To detect license plates, the cameras must be placed at a 45° angle to the road, pointing towards the direction of travel, so they get a good view of passing vehicles' rear plates (Figure 7). We also found that cameras at a 90° angle to the road were superior for calculating the speed of the vehicles that pass them, as the steeper angle made the motion of the vehicles clearer (Appendix D). We did note that cameras are very simple to reorient

or relocate compared to Eco-Counters, which are semi-permanently installed in the ground.

Figure 7:

Differences between different camera angles from license plate reading.

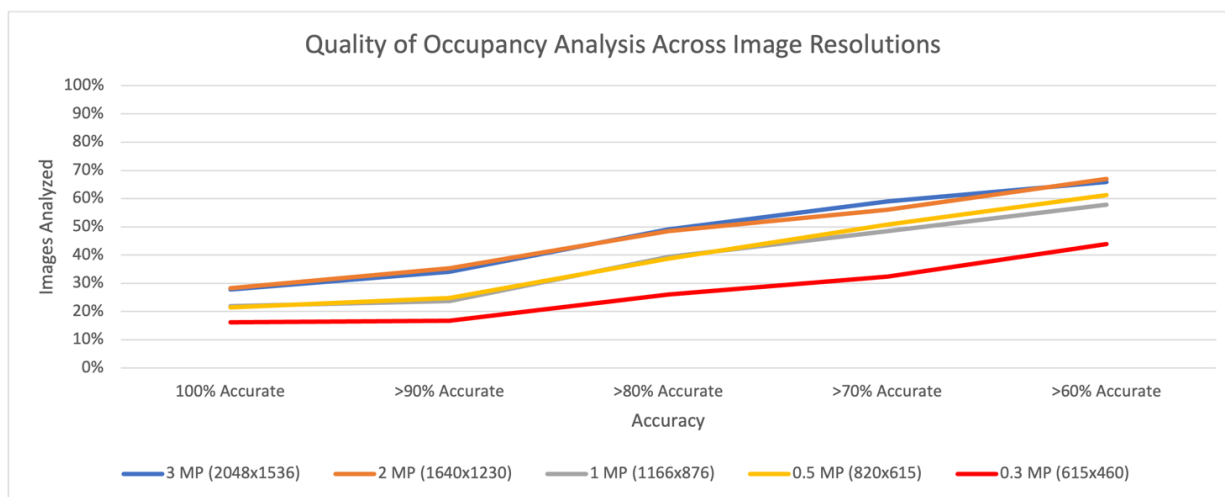


We tried to strike a balance between delivering sharp, detailed images and minimizing power consumption and storage space. We tested different image resolutions by running the same set of images through our algorithm multiple times, decreasing the resolution each time and comparing detection accuracy. We settled on using 3 MP (megapixels) for our tests since all the cameras we used provided this resolution and it performed well on initial detection tests. We performed this test on a set of 173 images from a camera on the rocks near Bass Harbor Head Lighthouse since that camera was already far away from the people being observed, so each entity took up a small area of the image to begin with. As shown in Figure 8, reducing image resolution too far has a profound impact on output quality, thereby reducing the algorithm's accuracy. Given that the results for 2 MP and 3 MP are very similar, increasing the resolution further is unlikely to improve quality. The lowest resolution we tested, 0.3 MP, is significant since the cellular cameras

we used only upload images at this resolution. These low-quality images could lead to inaccuracies in data output and misinterpretation of the information being processed, making it essential to set the right resolution.

Figure 8:

Comparison of image resolutions on algorithm results, run on 173 images from the Bass Harbor Head Lighthouse rocks.



We found that the battery performance of the cameras was a limiting factor on all the cameras that we tested. Recording video depleted the battery of our cameras in less than five hours of continuous use. Some of the cameras had solar panels, but they struggled to power the cameras due to being strapped to trees whose canopy left them in shadow. The cellular cameras used more power than the others to transmit their images over the network.

4.3 Computer Vision Meets the Needs of Acadia National Park

Our algorithm testing yielded a variety of statistics indicating the algorithm's accuracy compared to ground truth data. All the goals that we performed tests on had at least one algorithm provide sufficiently accurate results, as shown in Table 3. Some of the algorithms only work on photo or video data, while others work on both.

Table 3:*Comparison of the efficacies of all the algorithms*

Goal	Algorithm	Media Type	Efficacy*
Entity Identification	Entity Classification	Photo	89%
		Video	
Direction	Pose Estimation	Photo	94%
		Video	94%
	Frame Track Analysis	Video	>99%
Occupancy	Overview Detection	Photo	87%
	Exit/Entry Detection	Video	98%
Dwell Time	Vehicle Reidentification	Photo	<25%
		Video	
	License Plate OCR	Photo	93%
		Video	
Speed	Frame Track Analysis	Video	93%

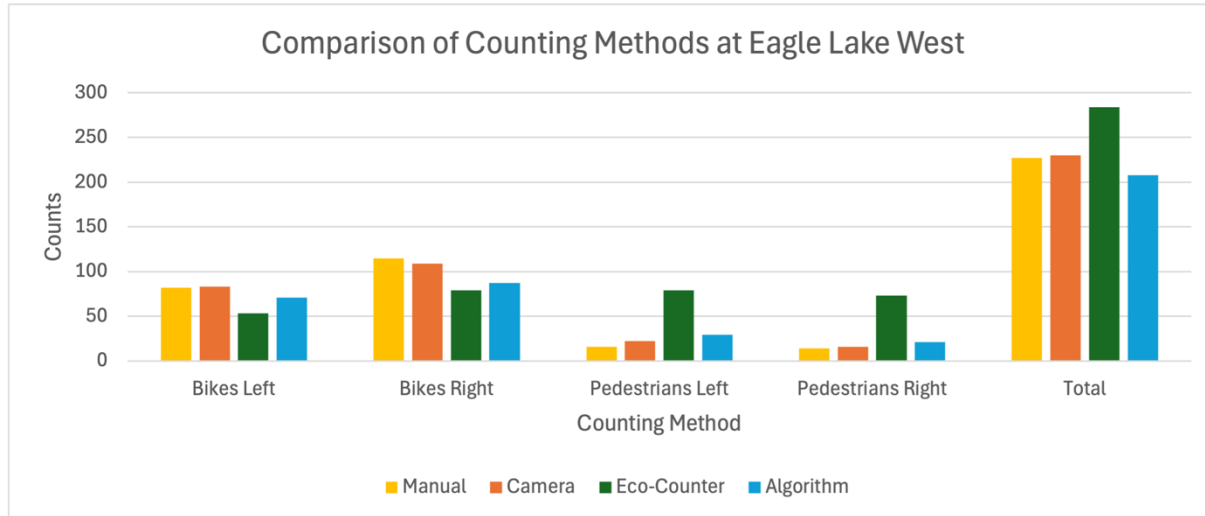
Note. Each algorithm was preprocessed with a YOLOv26 model to isolate entities of interest for further processing.

* The exact metric of efficacy varies between each algorithm and are not directly comparable.

Entity identification is the foundation of the more involved algorithms we implemented according to the team's requirements. To be able to make any inferences about more complex visitor behavior, we needed to be able to count and identify each passing entity, so we knew what to count it as. To benchmark this, we compared our entity identification to manually collected data, as well as Eco-Counter data looking across the Eagle Lake carriage roads over a two hours period. We found that the algorithm was 89% accurate when counting bikes compared to the manual count of 197 bikes and was more accurate compared with the Eco-Counters, which were counted 67% of the bikes, as shown in Figure 9. Additionally, our algorithm was able to differentiate between far more types of objects than Eco-Counters and car counters, including people, bikes, pets, and cars.

Figure 9:

Comparison between manual, camera, and Eco-Counter counts at Eagle Lake West, after merging bikes with riders



One major limitation of our detection algorithm that we ran into is that it counts people on bikes and bikes themselves as separate entities, which is not ideal in cases where pedestrians need to be differentiated from people on bikes. To account for this, we added a feature to our algorithm to combine bike and person entities with significant overlap into a single bike entity on the assumption that because they are mostly in the same place, the person is likely sitting on the bike.

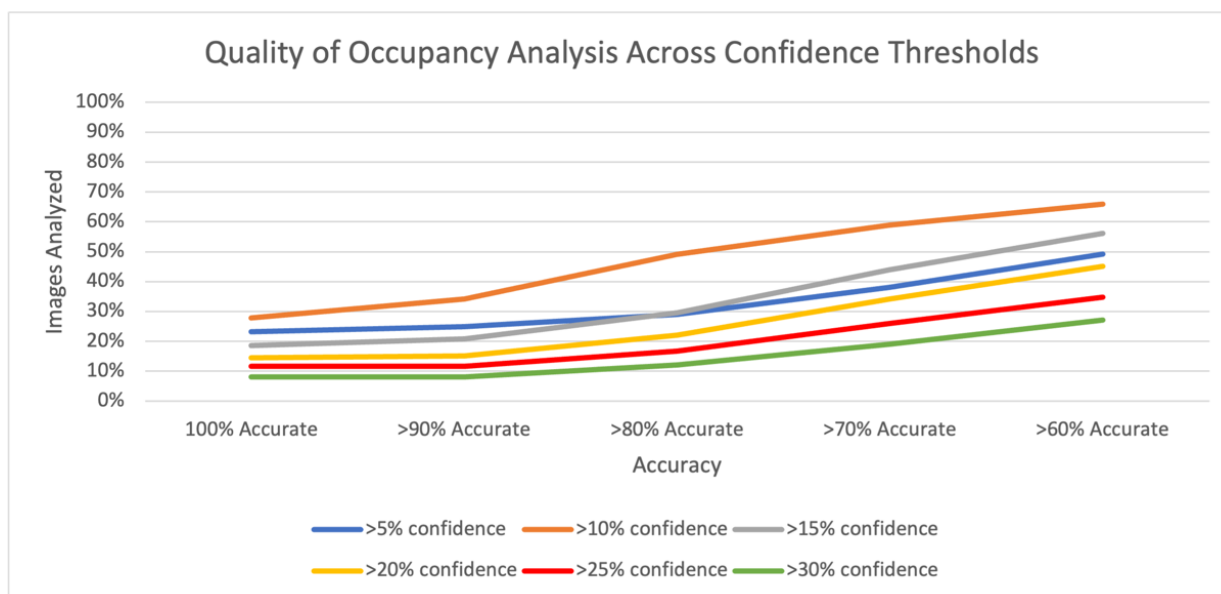
Alongside identification, determining the direction entities are moving is foundational to several statistics the park and teams were interested in. We used two different algorithms to determine the direction of entities; one used time-based direction detection on videos, and the other was designed to work on still images. Using a set of manually labeled videos and images, we found that the algorithm for detecting direction on video footage was 100% accurate on a dataset of 3 hours of footage, featuring 118 entities including pedestrians, cars, and bikes. The algorithm for still images was 94% accurate when tested on the same types of entities when using pose-based direction analysis.

After identifying entities in images, we were able to develop two separate algorithms to determine the occupancy of an area: one counted the number of entities visible to an

overview camera and the other used entrance and exit monitoring. The occupancy algorithm which used overview cameras was benchmarked by comparing the number of entities manually counted in each image to the number of entities identified by the algorithm in those images. The algorithm produced a confidence value indicating how certain it is of a given piece of data, which we used to filter out uncertain results. Figure 10 shows that for this set of images, results were best when ignoring anything with confidence below 10%. With this threshold, 87% of the 173 images were identified by the algorithm to have counted within three people compared to the manual count. This algorithm faced the same limitations as the detection algorithm.

Figure 10:

Comparison of occupancy analysis quality across confidence thresholds, from 173 images from the Bass Harbor Head Lighthouse rocks.



The entry/exit occupancy algorithm was benchmarked by comparison of manually counted entities passing in and out of an area. Using two hours of video at the Jordan Pond parking lot entrance, we found that our algorithm was able to accurately count 98% of total passing entities over a 2.5-hour period featuring approximately 200 entities.

We attempted to create a clustering algorithm which used the positions of entities and their size to group them together. This algorithm turned out to be incapable of accurately grouping entities together in all but the simplest cases. We qualitatively observed that it had extremely low accuracy when running on images of crowds at Bass Harbor Head Lighthouse. It frequently failed when people were passing near each other but not part of the same group and when the overall density of people in the image was high.

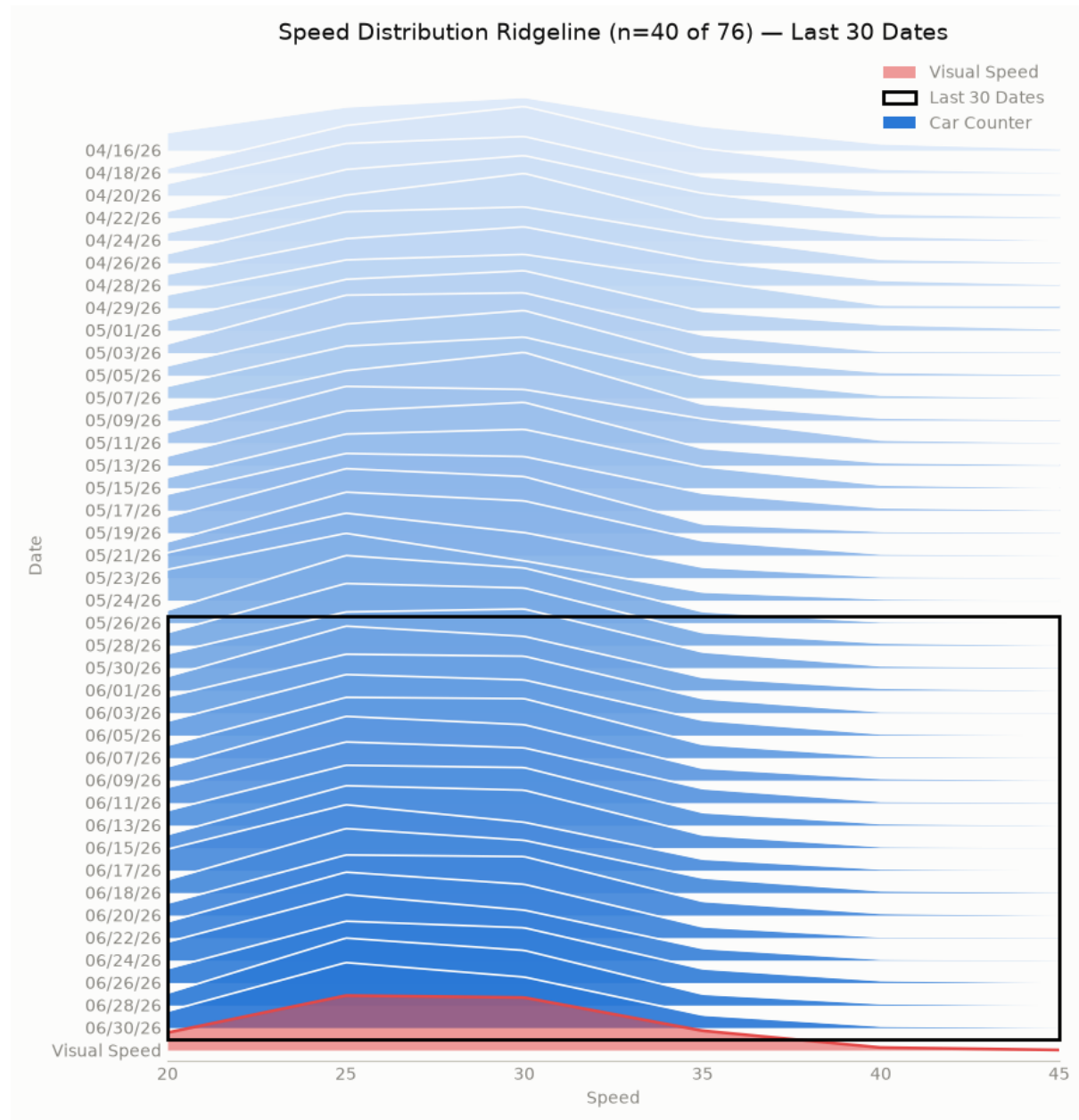
In addition to occupancy and clustering, the Jordan Pond, Schoodic Peninsula, and Bass Harbor teams were all interested in the dwell times of park visitors. We first attempted to calculate dwell time statistics using fingerprint reidentification techniques, but we quickly found that differences in image backgrounds, motion blur, and differences in entity orientation made this reidentification method inaccurate with our data. We switched to performing dwell time calculations on vehicle license plates, of which our algorithm could read 92% of over 7000 images using a smartphone camera. Furthermore, the algorithm achieved over 98% accuracy when reading individual characters. Over 67% of the license plates that were missed were only missed by one character, which means fuzzy matching is most likely able to overcome these misses. We were limited in benchmarking the accuracy of dwell-time calculation because of camera and dataset limitations. The cameras were only able to capture a few hours of footage at a time, which limited the number of vehicles that entered and exited during the recording period, and time-sync issues with the cameras prevented accurate time measurement. Benchmarking dwell time properly will likely reveal other limitations and bugs with the algorithm itself, but overall, we feel it is a good start.

The Schoodic Peninsula group and the e-bikes group expressed an interest in determining the speed of passing entities. The historical data from car counters located around the park provided the distribution we needed to compare against, but we were unable to access the ground truth data that matched the time we collected our data. We found that our data passed a Jensen-Shannon divergence test on the month of car counter data immediately preceding it with a score of 0.021 and a 93% overlap, which means the distribution of car counter data did not change shape drastically over the course of a

month. Figure 12 shows the result of this test and the distribution of speed over time from the camera data.

Figure 12

Comparison of speed distributions for car counter on Park Loop Road north of Jordan Pond



Note. The blue distributions closely match each other showing little deviation over the course of 30 days, indicating that period's viability as a ground truth for testing our camera speed algorithm.

5. Recommendations

Based upon our findings, we have identified three areas that the park could work on toward achieving the goal of more efficient VUM data collection:

1. Continuing the development and evaluation of analysis software to simplify the usage of computer vision algorithms.
2. Investing in specialized camera hardware suitable for reliable video capture, real-time connectivity, and onboard processing.
3. Improving the computer vision pipeline and implementing specific tasks such as differentiating between e-bikes and traditional bikes.
4. Perform more robust benchmarking procedures to evaluate algorithms in a wider range of scenarios.

In this section, we explain how our findings justify the need for these new developments and cover important considerations when making these improvements.

5.1 Analysis Software

As part of our development of the proof-of-concept data collection system, we have created a comprehensive user interface to simplify the use of our algorithms. This interface provides access to all our algorithms without requiring significant technical expertise and allows for exporting into other data analysis tools. We utilized this program extensively during our testing and believe that it could be used and improved by the park and other research teams. A user manual for this software is in Appendix E, and the source code is available for improvement.

This software is still in the early stages of development, and to continue to be used, it should undergo extensive benchmarking to ensure that it provides accurate metrics and meets the needs of the park. It should also undergo user testing to make sure that researchers are able to make use of it effectively. Future changes to the data collection system, such as the addition of real-time remote data access, will also require changes to the software.

5.2 Specialized Camera Hardware

Our assessment of the cameras currently used in the park and those we used for testing revealed that they lack many of the features needed for an end-to-end data collection system. We have evaluated the issues that current cameras face and determined the most important criteria when selecting or designing a camera for this purpose. We recommend that the park invest resources in finding or developing cameras with these features to support more advanced computer vision systems.

Throughout all the tests we conducted, the most significant issue was the inability of the cameras to capture photos or video at a frequent and predictable rate. The computer vision algorithms we worked with required images to be collected frequently to track the movement of an entity across multiple frames. Most of the usable data that we collected came from videos, so we recommend that a camera designed for use with computer vision be able to record video for the entire duration motion is detected.

Networking capabilities need to be added to cameras that are used in a remote system. Remote real-time data collection is a promising next step for this technology, but existing cameras faced unreliable connections and uploaded low-resolution images. A camera capable of quickly and reliably uploading high-resolution images or video to the internet would be required for real-time analysis.

Instead of uploading high-resolution copies of every image to the internet, it is also possible to run the computer vision models onboard the camera itself. This would significantly reduce the network requirements for a remote data collection system as the camera would only need to upload the statistics that it gathered from each photo or video rather than the entire file. This would also improve the privacy measures of the system; images would be deleted as soon as statistics are gathered and would never leave the camera. However, this would likely increase the cost of the camera because it would need more advanced hardware to run the algorithms on the camera. We feel that the benefits outweigh the drawbacks of this option and recommend it as a means of achieving real-time remote data collection in the absence of improvements to networking infrastructure in Acadia National Park.

5.3 Continued Computer Vision Pipeline Development

There are some tasks that the park and the research teams were interested in that we were unable to design an algorithm to complete, such as differentiating e-bikes and regular bikes and differentiating clusters. Additionally, it is likely that automating additional tasks will be one of the next steps if the park does use our program. We recommend that the park have technically inclined volunteers upgrade our computer vision system to align more closely with their needs.

E-bike detection is a particularly difficult challenge because there are no preexisting computer vision algorithms to base the system on. Only manual observation is currently capable of accurately counting e-bikes, so developing a computer vision system which can detect them would be valuable. From our observations on the carriage roads, it is usually possible to visually differentiate between the types of bikes; a specially trained computer vision model could likely pick out differentiating factors with sufficient training.

Our clustering model, while effective in some scenarios, proved to be inaccurate in most use cases. A future addition to this tool suite should be to overhaul our algorithm implementation with a much more accurate one. Several benchmarking datasets can be used to quantify this new algorithm's accuracy. We also could not effectively test the dwell time algorithm due to limitations with our cameras and dataset. A future addition to this project would be continued development on this algorithm to be able to track dwell time of vehicles in parking lots and other similar scenarios.

5.4 Improve and Test New Benchmarking Procedures

Overall, our system performed well and met most of our goals on the datasets we generated from footage from the national park, but there are still some algorithms that need improvement, and the data we collected was not very diverse. We could not collect data on days of inclement weather, which limits the efficacy metrics' credibility. A future addition to this project should be testing the algorithms' performance on data collected during alternative weather conditions, as well as other areas of the park than the ones we

collected from. This will hopefully inform future improvements to our algorithms such as dwell time and clustering, as well as make the system more representative.

5.5 Conclusion

Our findings indicate that computer vision is a viable option for data collection in Acadia National Park. We had successes using computer vision for entity identification, counting, occupancy measurement, dwell time, speed estimation, and direction determination with images collected throughout the park. Our benchmarks found that these methods produced similar results to manual observation and outperformed the park's existing automated data collection methods in some cases.

Computer vision is a key component toward improving Acadia National Park's automated data collection systems. Combined with specialized camera hardware and custom models, computer vision can enable efficient automated VUM data collection, allowing the park to gather more data with less effort. This will allow the park to respond more effectively to current congestion challenges along with other issues that may arise in the future. Analysis would require less time reviewing images, leaving more chances to make actionable improvements to the park. Investing in computer vision and data collection in general will allow Acadia National Park to keep up with the growing demands of today's tourists and continue running smoothly for years to come, and the tools that we developed pave the way toward the systems needed to do so.

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Glossary

Algorithm: A step-by-step procedure or a set of rules designed to solve a problem or perform a task.

Benchmark: Standardized tests that are conducted on a model or system to assess and compare its performance with other models or baselines.

Bounding Box: The tightest possible rectangular frame wrapped around an object in an image or video.

Car-Counter: A tool that counts and measures the speed and direction of passing cars.

Cellular Network: The type of wireless network used by cell phones among other devices.

Computer Vision: A type of algorithm that allows computers to analyze images like humans, converting visual data into quantitative statistics autonomously.

Clustering: The process of grouping entities based on their distance.

Dwell Time: The measure of how long an entity stays in a certain area.

Eco-Counter: A proprietary data collection tool that counts passing pedestrians and bikes.

Entity: A specific object of interest.

Field of View: The area that a camera can see at once.

Fuzzy Matching: Matching similar pieces of text when a few characters are missing.

Frame: An image extracted from a single point in time in a video.

Frame Track Analysis: The process of analyzing a video one frame at a time, tracking the movement of an entity across frames.

Jensen–Shannon Divergence Test: A calculation that measures how similar two distributions are.

Machine Learning: A field of data science in which computers iteratively improve their ability to analyze data.

Occupancy: The number of people in an area over time.

Optical Character Recognition (OCR): An algorithm that extracts writing from images and converts it into editable, searchable digital text.

Pose Estimation: The process of estimating the position and orientation of an entity in an image.

Reidentification: The process of matching entities to previous times they have appeared.

Resolution: The number of pixels in an image.

Visitor Use Management (VUM): A framework to ensure resource protection and high-quality visitor experiences on federally managed lands and waters.

You Only Look Once (YOLO): An object-detection algorithm that identifies and locates objects in images or videos.

Appendices

Appendix A: Park Staff Interview Guide

Thank you for agreeing to participate in this interview. My name is _____ and my team and I are working with Acadia National Park and Friends of Acadia to determine the feasibility of using computer vision to aid VUM data collection in the park. Our findings will be used to inform the design of a proof-of-concept data collection system and will be documented as part of our process and published in a research paper.

The purpose of this interview is to understand what Acadia currently does to monitor visitors. This interview will take approximately twenty minutes and is entirely voluntary; you may skip questions or stop the interview at any time. We would like to record this interview with your permission. The recording will not be shared and will be deleted after the research project. Your responses will be considered alongside other participants' responses to assess which locations are most suitable for placing cameras. Only the results of our analysis will be published; your name and other identifying information will not be publicized. If you have any questions before we get started, please ask them now.

For more information about this research, the rights of research participants, or in case of research-related injury, contact:

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1. What is your role in the park?
2. What is your experience with VUM data collection as it stands?
 - a. Do you go out into the field to collect data?
 - b. How do you process the data after it is collected?
 - c. How long do you spend working with the system per month/week/day?
 - d. Is there any part of your process that feels very streamlined?

- e. Are there any parts that feel awkward?
- f. What takes the longest?
- g. Do you work with anyone else when you do this?
 - i. How many other people?
- 3. Which areas of the park do you perform data collection in?
 - a. How long do you spend with the data from that area?
 - b. Is there anything about that area which is particularly inconvenient?
- 4. What would make working with visitor data easier?

Appendix B: Computer Vision Data Collection Privacy Statement

All images collected in this study will only be for counting visitors and analyzing general visitor movement trends. Images will not be used to identify any individuals or gather personal information about them. Facial recognition and any other methods to determine personal characteristics, such as age or gender, will not be used. All data will be handled to keep individuals' information private and anonymous and will be deleted at the conclusion of the study.

Appendix C: Observation Protocols

The purpose of manually observing the locations the cameras are in is to gather control data to benchmark the cameras. The protocol for observation benchmarked against different control data methods is outlined below:

Counting by Eco-Counters

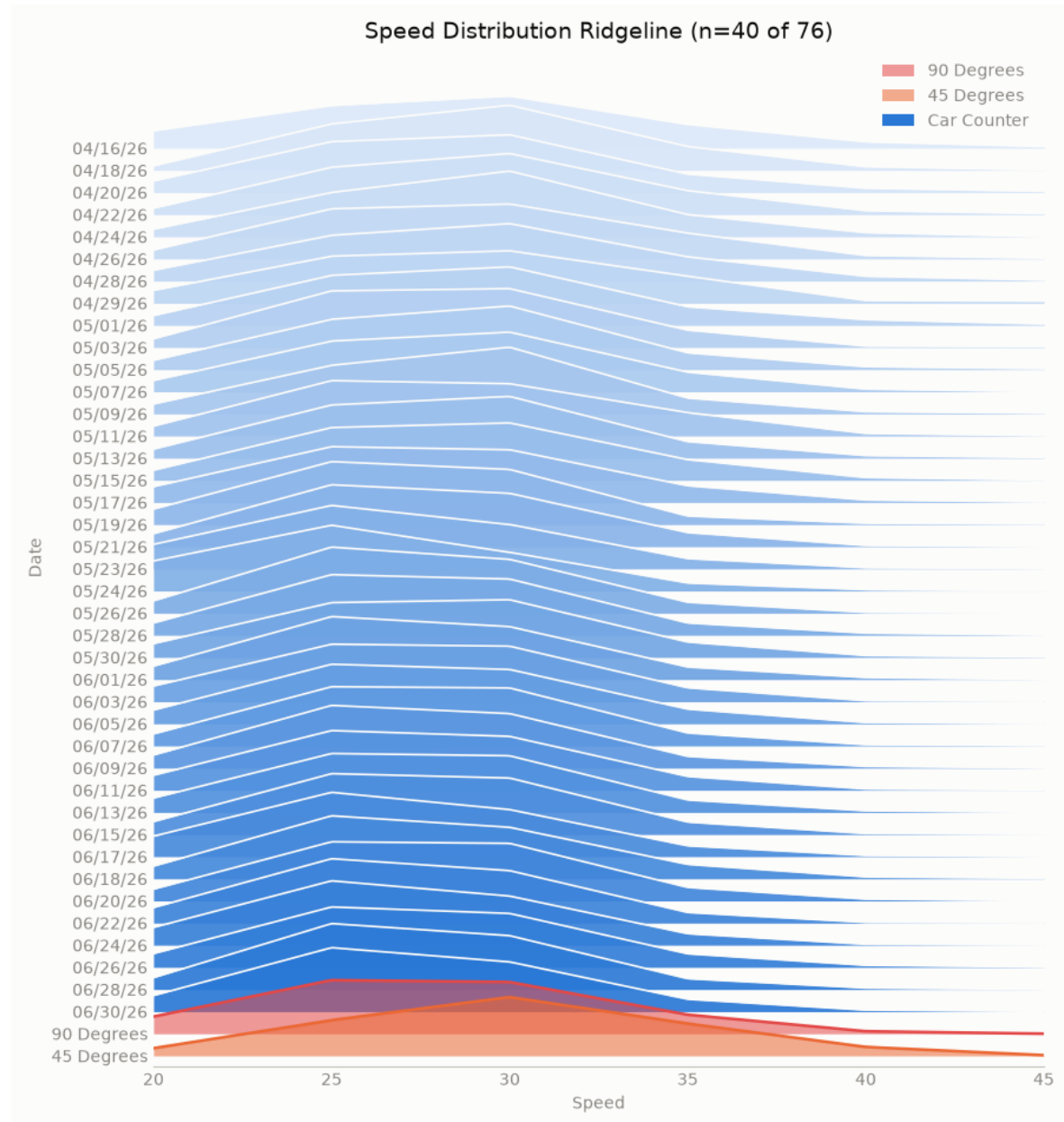
- Set up the camera so it sees the Eco-Counter measurement location
- Verify the camera's field of view and that it is recording images or video as intended
- Begin counting the entities that pass by the camera and Eco-Counter
 - Record counts in 15-minute intervals
 - Record entity type and direction
- Collect Eco-Counter and camera data
- Finish counting at the end of one of the 15-minute intervals
- Count the entities the camera saw, using the same methods as in-person counting
 - Record counts in 15-minute intervals
 - Record entity types and directions

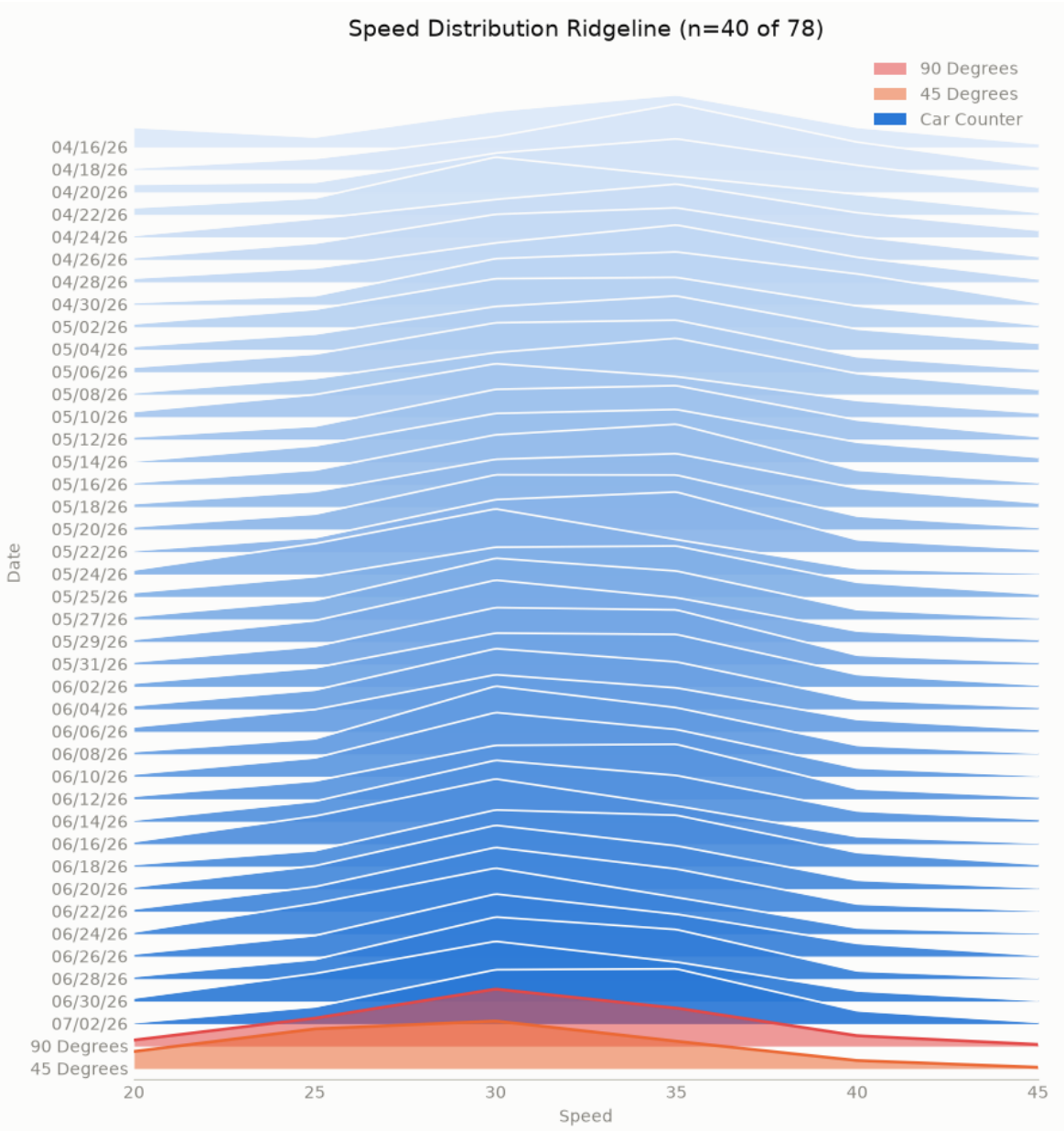
Counting by Car Counters

- Set up the camera in view of the car counter measurement location
- Verify the camera's field of view and that it is recording images or video as intended
- Record video for as long as possible
- Count the entities the camera recorded
 - Record times each entity passed the camera
 - Record entity types and direction

Appendix D: Speed Estimation Camera Orientation Comparison

Speed estimation algorithms using cameras can be sensitive to camera angles. The following figures illustrate that the 90° cameras perpendicular to the road have higher accuracy than 45° orientations.





Appendix E: Analysis Software User Manual

Installation and Setup

To install the program, go to <https://github.com/ElliotScher/Acadia-AI-E26/releases> and download the executable for your system. You may wish to place the executable in your applications folder for easy access.

When running the program for the first time, your system or antivirus software may warn you about potential malware. This can safely be ignored. If you are concerned about malware, the source code for the program is available for review on GitHub at the link above. It may take up to 15 seconds for anything to appear while the computer vision pipeline is initializing.

Importing Images and Videos

When importing media from a camera, it is recommended to transfer all media off the camera's storage onto your device to speed up processing. To import images or videos, click on the **File** menu and select **Open Image Folder** or **Open Video Folder** depending on the type of media you are working with. If the images or videos are spread across multiple subfolders, select the folder which contains all of them. For example, in this case the **eagle-lake-images** folder should be selected, which will grab all the images from the subfolders.

- eagle-lake-images
 - 2026-07-01
 - 07-48-02.jpg
 - 07-48-12.jpg
 - ...
 - 2026-07-02
 - 07-32-49.jpg
 - 07-33-01.jpg
 - ...
 - ...

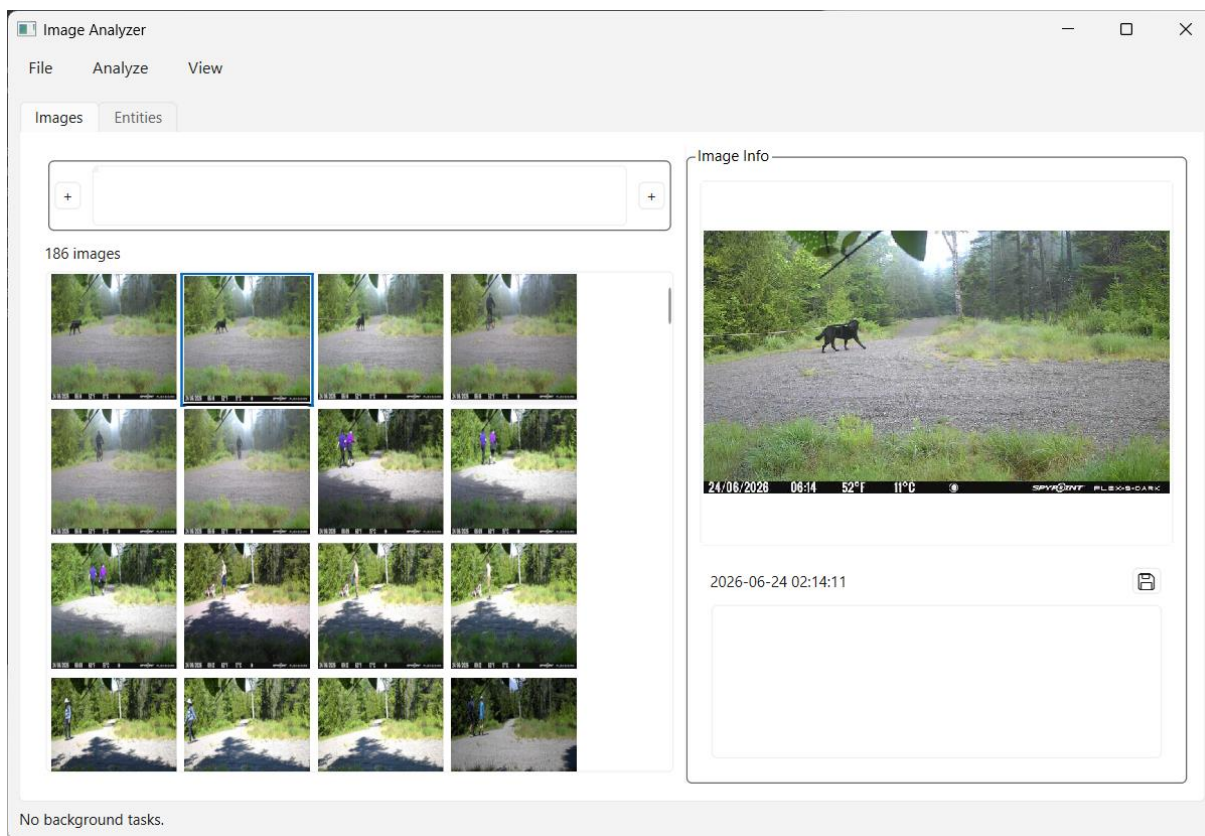
When importing images, all detected images will be listed in the **Images** tab. You can click on an image on the left to enlarge it on the right. Once an image is opened on the

right, the image can be panned by scrolling or using the scroll bars. It is possible to zoom in the right panel by holding the control key while scrolling or use the zoom controls in the **View** menu.

When importing video, you will be taken directly to the analysis screen to extract key frames from the video, which can then be manipulated similarly to imported images.

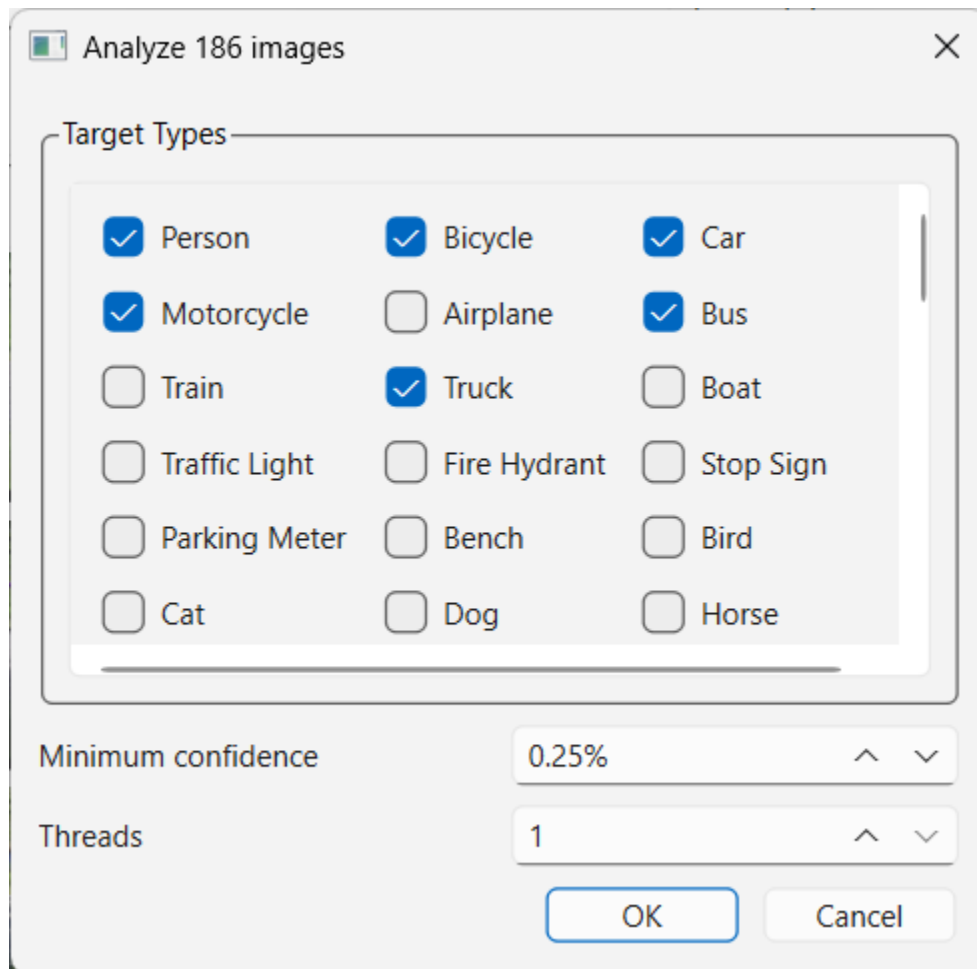
The program will create a file named **photos.db** or **videos.db** in your selected folder to store the results of analysis. When reopening previously analyzed media, select the folder where this file was created.

Running Analysis Tools



Most of the analysis tools can be run either on everything by specifying **All** or only on a filtered subset of the media by specifying **Filtered** (see the **Filtering Results** section). Some also provide a **Threads** option, which controls how many tasks your computer will work on at once. Setting this to a higher number will speed up analysis but may slow down

other programs running on your computer. While analysis is running, progress can be monitored in the progress bar at the bottom of the window.



The base analysis must be completed before using other analysis features. This is the first option in the **Analyze** menu and will appear automatically when analyzing videos. This will prompt you to select which types of entities you are interested in analyzing.

The **Minimum Confidence** option

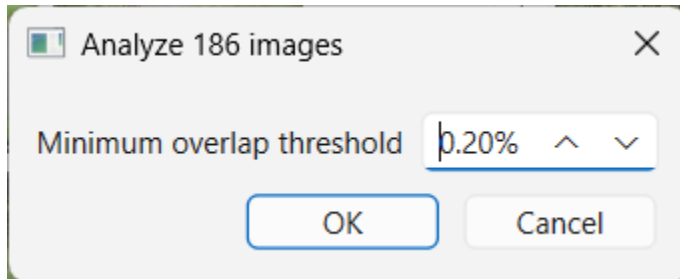
The **Exclusion Margin** option allows you to

The **Merge Similar Entity Types** option treats similar entity types as a single type for easier analysis, such as buses and trucks → cars, and motorcycles → bikes.

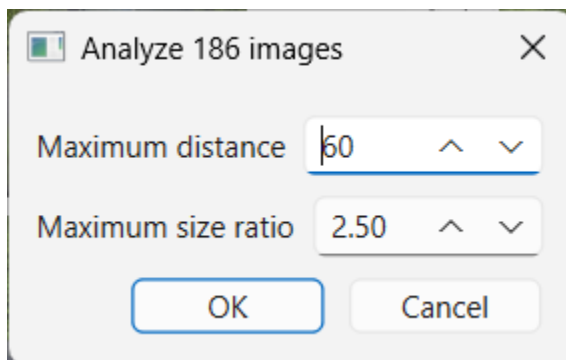
When processing video, the **Downsample Factor** option controls how many video frames to skip between each analysis. Lower values provide more detailed results but take longer to process. The **Variable Downsample Factor** option controls how many frames to skip dynamically when previous frames are empty.

Once you have run the base analysis tool, you will be able to access the **Entities** tab, which lists individual entities identified in your media. You will also be able to choose from several other tools depending on the data you wish to look at.

- **Merge Bikes and Riders** – Merges bikes and their riders into singular entity based on their overlap. Allows you to set a minimum overlap threshold before merging can occur. If using with frame-by-frame tracking, do this first.

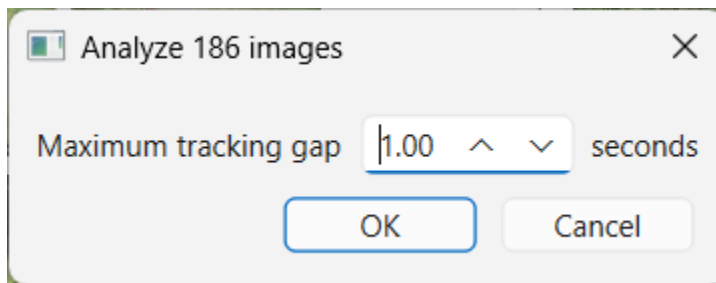


- **Find Clusters** – Groups entities in an image into clusters. Allows you to set the maximum distance in pixels and the maximum ratio of sizes for two entities to be clustered together. Once this tool has been used, entities will list other entities that they are clustered with in the **Entities** tab. Does not work well with frame-by-frame tracking.

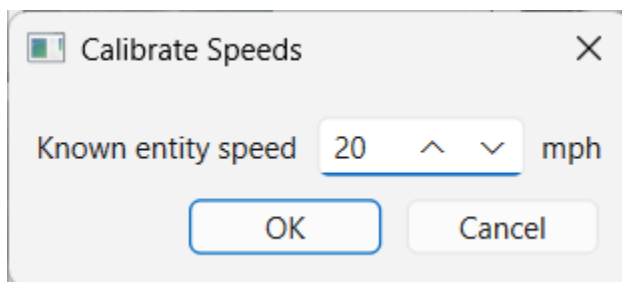


- **Frame-By-Frame Tracking** – Tracks the movement of entities across multiple images. Works best with images taken in very quick succession, such as frames extracted using the video import tool. Allows you to set the maximum time that an entity will continue to be tracked if it disappears temporarily. Once this tool has been used, entities in the **Entities** tab will list all images that they appear in and will

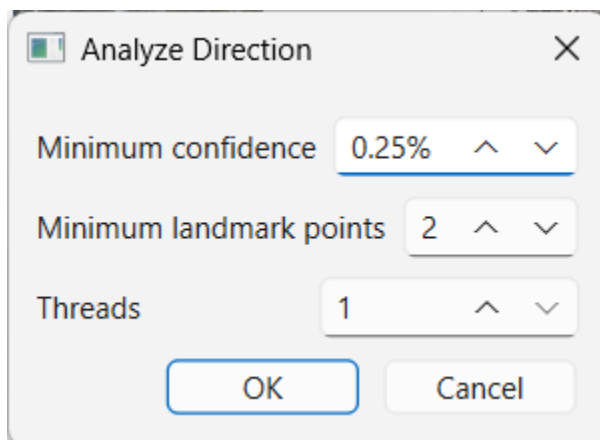
be populated with their left/right travel direction.



- **Calibrate Speed** – After using the frame-by-frame tracking tool, use this tool to calibrate movement speeds based on the speed of a known entity. To use this tool, go to the **Entities** tab and select an entity that you know the speed of. Then select the **Calibrate Speed From Selected Entity** tool and enter the known speed of that entity. All other entities' speeds will be calibrated relative to that speed.



- **Direction From Pose** – Determines the movement direction of person entities from still images based on the position of their body. Allows you to set the minimum confidence value for a pose to be considered and the minimum number of pose landmarks needed for a direction. This tool will populate entities with both left/right and forward/back directions (forward is towards the camera, back is away).



Filtering Results

The program provides several filters to aid in sorting through the analyzed images and entities. Filters can be applied using the bar located above the list of images/entities. All filters in the same row must be satisfied for images to be shown. The + button on the left can be used to create additional rows which display images based on alternative conditions; images will be shown if they match at least one row. The – button on each filter can be used to delete it; the – button on the left of a filter row (other than the first one) can be used to delete the entire row.

- General Filters
 - **Time Range** – Filters images/entities that appear within a certain time range. Uses 24-hour time formatting. For entities, only matches entities that fully exist within the time range.
 - **Date Range** – Filters images/entities that appear within a certain date range. Uses year-month-day date formatting. For entities, only matches entities that fully exist within the date range.
- Image Filters
 - **Analyzed / Not Analyzed** – Filters images that have/have not been through the base analysis pipeline.
 - **Entity** – Filters images based on the type and number of entities that appear in the image. Click on the > button to expand show additional options. Also allows filtering by the minimum confidence value to count entities by. Only works once the image has been analyzed.
 - **No Entity** – Similar to the **Entity** filter, filters images based on the lack of appearance of a type of entity.
 - **Cluster Count** – Filters images based on the number of clusters that appear in the image. Only works once clusters have been found.
- Entity Filters
 - **Type** – Filters entities based on the type of entity

- **Cluster Size** – Filters entities based on the size of the cluster that they appear in. Only works once clusters have been found.
- **Count / Confidence** – Filters entities based on the number of times that entity appears and the minimum confidence values of each appearance.
- **Direction** – Filters entities by their direction of travel. Only works once pose directions or frame-by-frame tracking has been run.
- **Speed** – Filters entities by their speed of travel. Only works once frame-by-frame tracking has been run and speeds have been calibrated.

Exporting Results

To export the results of the analysis, click on the **File** menu and select **Export > Filtered** or **Export > All**. The program provides four different exporting modes, all of which create CSV (comma-separated values) files. These files can be imported into spreadsheet such as Excel. Note that some information such as speed, direction, and clusters will only be populated in the export if the appropriate analysis tool has been run.

- **Row per Image** – Each image captured is represented by a row in the spreadsheet and includes the date and time of the image, the number of each type of entity detected in the image, optionally separated by direction, and the number of clusters detected in the image.
- **Row per Entity** – Each entity detected is represented by a row in the spreadsheet and includes the start and end date and time that the entity appeared, the type of entity, the number of other entities clustered with it, and the speed and direction it was travelling.
- **Row per Cluster** – Each cluster detected is represented by a row in the spreadsheet and includes the start and end date and time that the cluster appeared and the number of entities in the cluster. Only works once the **Find Clusters** tool has been used.

- **Row per Time Interval** – Similar to Row per Image, but the results from all images taken within a given time interval are aggregated. When this option is selected, a box will appear to enter the length of the time interval to group images by.

The **Separate by Direction** option, only available for the **Row per Image** and **Row per Time Interval** options, controls whether separate columns are created to count entities travelling in each direction. Note that since an entity may be tagged with multiple directions, the total of the counts may be greater than the total number of entities.

The **Open When Exported** option will automatically open the exported file once the export has completed. This option will use the default application set to open CSV files, which is often Excel if it is installed on your system.

Appendix F: Algorithm Benchmarking Procedures

Entity Identification

1. Run default counting procedure
2. Export results to CSV
3. Plot results against manual counting

Occupancy

Entry/Exit

1. Run default counting procedure
2. Replace manual counts with identified counts
3. Run entry/exit counting procedure
4. Compare results against manual counting

Overview

1. Manually count entities in each frame
2. Run overview algorithm
3. Compare results

Dwell Time

1. Run license plate detector
2. Discard human unreadable plates
3. Label remaining license plates
4. Run license plate OCR
5. Compare results against manually labeled plates

Direction

1. Manually determine entity directions
2. Run direction algorithm on videos
3. Compare results against manual directions

Speed

1. Aggregate car counter data
2. Run speed algorithm with calibration speeds
3. Compare Jensen-Shannon divergence score